Four-Packet Gain Floors and a Source-Level Lower-Bound Attack

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Abstract

The previous TPC-276 compiler identified the signed packet gain $r = D/G$ as a possible source of endpoint margin. This paper isolates what four-packet geometry can actually provide. For arbitrary four vectors, $G \le 4D$, hence $r \ge 1/4$; if the net signed cross term is nonpositive, then $G \le D$ and $r \ge 1$. Writing $\kappa = (D - G)/D$ gives the exact inverse coordinate $r = (1-\kappa)^{-1}$, so a polynomial gain requires near-total cancellation, not merely a favorable sign. An exact rational replay of the literal TPC source at eight registered and extended scales finds negative net cross term and $r > 1$ on every row, while the smallest row has $r < 1.01$. The result is a reusable geometric floor and a finite source diagnostic, not an asymptotic power gain or a twin-prime proof.

1 Question and frozen object

TPC-275 retained the four actual consecutive source-block packets V_0, V_1, V_2, V_3 of the projected literal V59 operator. TPC-276 showed that their signed energy can improve a margin, but left open whether the ratio has a source-level lower bound. We first remove an ambiguity: which part of such a lower bound is forced by geometry, and which part must come from arithmetic structure?

Throughout, the source, prime shell, masks, deleted diagonal, beta weights, and rank-three four-block Haar projection are exactly those frozen in the TPC-268–275 chain. No synthetic packet is introduced. The new finite rows are a declared extension of the same computation; they are not a claim about an already proved asymptotic registry.

2 The geometric floor

Let the inner product be linear in either convention and take real parts in the complex case. Define

$$D = \sum_{j=0}^{3} \|V_j\|^2, \qquad S = \sum_{j=0}^{3} V_j, \qquad G = \|S\|^2,$$

and

$$E = \sum_{0 \le j < k \le 3} \mathrm{Re}\langle V_j, V_k \rangle.$$

Expansion gives $G = D + 2E$.

Theorem 1 (universal four-packet floor)**.** *If $D > 0$ and $G > 0$, then*

$$G \le \left(\sum_{j=0}^{3} \|V_j\| \right)^2 \le 4D, \qquad r := \frac{D}{G} \ge \frac{1}{4}.$$

If in addition $E \le 0$, then $G \le D$ and $r \ge 1$.

N	H	Q	z	$r = D/G$ (display)	κ (display)
192	32	6	5	1.006248221	0.006209423
256	38	6	6	1.058743787	0.055484422
384	50	7	7	1.124746633	0.110910875
512	64	8	7	1.080616361	0.074602203
768	86	9	9	1.051461797	0.048943098
1024	108	10	10	1.054717352	0.051878688
1536	150	12	12	1.018190110	0.017865141
2048	170	12	12	1.030357019	0.029462622

Table 1: Exact source replay, kernel exponent $s = 2$. Decimal values are display-only; threshold decisions use outward rational intervals.

Proof. The first inequality is the triangle inequality followed by Cauchy–Schwarz in $\mathbb{R}^4$: $(\sum_j \|V_j\|)^2 \le 4\sum_j \|V_j\|^2$. The signed assertion follows from $G = D + 2E \le D$. Positivity of G permits division. $\square$

The constants are sharp. Four equal aligned vectors attain $r = 1/4$. Four nonzero mutually orthogonal vectors attain $r = 1$. Thus even the stronger signed floor is only a constant statement.

3 The cancellation coordinate

Define the dimensionless cancellation fraction

$$\kappa = \frac{D - G}{D} = -\frac{2E}{D}.$$

Whenever $G > 0$, the preceding expansion is equivalent to

$$\frac{G}{D} = 1 - \kappa, \qquad r = \frac{1}{1-\kappa}. \tag{1}$$

For a nonpositive cross term, $0 \le \kappa < 1$. Formula (1) identifies the actual missing theorem: a bound $r \ge bx^\gamma$ is equivalent to

$$\frac{G}{D} \le b^{-1}x^{-\gamma}, \tag{2}$$

or $1 - \kappa \le b^{-1}x^{-\gamma}$. Merely proving $E \le 0$ gives $\kappa \ge 0$ and no positive power. An orthogonal family has $\kappa = 0$ for every scale and is an exact adversary to geometric power promotion.

Proposition 2 (no geometric power credit)**.** *The four-packet axioms alone imply no inequality $r \ge bx^\gamma$ with $b > 0$ and $\gamma > 0$ on a growing family.*

Proof. For every x, choose four orthogonal unit vectors. Then $D = G = 4$ and $r = 1$, contradicting $1 \ge bx^\gamma$ for sufficiently large x. $\square$

This proposition is not an assertion that the literal prime source is orthogonal. It says that an eventual source theorem must use information beyond the packet Gram axioms, such as a quantified arithmetic mechanism forcing G/D to decay.

4 Exact source scan

The certificate evaluates the actual beta source and prime shell with exact rational arithmetic. For each row it accumulates the four unprojected packet outputs, applies the three declared Haar contrasts to each output, and forms D and G. The JSON stores outward rational intervals on the grid 10^{15} and a digest of the exact pair; an independent checker uses column-major accumulation.

All eight rows have $E < 0$ and $r > 1$. The first row nevertheless has $r < 101/100$. Consequently this finite registry refutes the stronger one-percent floor as a scoped finite claim, while retaining the weaker negative-cross-term floor. The gain is also visibly nonmonotone: the sequence falls from $N = 384$ to $N = 768$ and again from $N = 1024$ to $N = 1536$. Neither observation supplies a growing quantifier.

5 Implications for the endpoint ledger

TPC-276 uses r through $m^2 = rm_D^2$. The present result sharpens the input firewall. A signed cross-term estimate can certify $r \geq 1$, but a positive exponent in the TPC-276 budget requires (2), which is a near-null estimate for the reassembled packet sum. The finite source rows are useful for locating that target and for testing implementations; they pay zero fixed-power credit.

6 Conclusion and route evaluation

The main positive result is the sharp, reusable four-packet floor together with an exact source replay at eight scales. The main obstruction is equally precise: geometry alone cannot provide a positive power, and the finite scan already fails a one-percent floor. The next natural experiment is to test cross-scale gain stability under controlled prime-shell and clock changes, then to formulate a source theorem directly for the deficit G/D.

`ROUTE-A`	not applicable to this finite gain-floor audit
`ROUTE-B`	yes, scoped source-gain floor and finite attack
`FIXED-POWER-CREDIT`	0
`ARITHMETIC-L2 / FULL-GATE-B`	open
`TWIN-PRIME-RESULT`	none

References

[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.

[2] L. Wang, “Signed Four-Packet Reassembly for the Literal V59 Output,” TPC-275 project release, 2026.